MLE-BENCH: EVALUATING MACHINE LEARNING AGENTS ON MACHINE LEARNING ENGINEERING

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ABSTRACT

We introduce MLE-bench, a benchmark for measuring how well AI agents perform at machine learning engineering. To this end, we curate 75 ML engineering-related competitions from Kaggle, creating a diverse set of challenging tasks that test real-world ML engineering skills such as training models, preparing datasets, and running experiments. We establish human baselines for each competition using Kaggle's publicly available leaderboards. We use open-source agent scaffolds to evaluate several frontier language models on our benchmark, finding that the best-performing setup — OpenAI's o1-preview with AIDE scaffolding — achieves at least the level of a Kaggle bronze medal in 16.9% of competitions. In addition to our main results, we investigate various forms of resource-scaling for AI agents and the impact of contamination from pre-training. We open-source our benchmark code (github.com/openai/mle-bench/) to facilitate future research in understanding the ML engineering capabilities of AI agents.

1 INTRODUCTION

Language models (LMs) have achieved impressive performance on many coding benchmarks (Chen et al., 2021; Hendrycks et al., 2021; Austin et al., 2021; Li et al., 2022) and are making progress on a variety of machine learning tasks, such as architecture design and model training (Zheng et al., 2023; Huang et al., 2024b). LMs have also been adopted into programming tools (Kalliamvakou, 2022), and progress in agent scaffolding has increasingly automated developer workflows (cognition.ai, 2024; Dohmke, 2024). However, while there has been a surge in development on model and agent capabilities, few benchmarks holistically measure autonomous end-to-end ML engineering.

We introduce MLE-bench, an offline Kaggle competition environment for assessing how well AI agents can perform difficult machine learning engineering (MLE) tasks. We built MLE-bench to be a robust measure of real-world progress in autonomous ML engineering agents, focusing on two main design choices: (i) selecting tasks that are challenging and representative of contemporary ML engineering work, and (ii) being able to compare evaluation results to human-level performance.

The resulting benchmark consists of 75 diverse Kaggle competitions across a variety of domains, including natural language processing, computer vision, and signal processing. Many of the competitions are contemporary challenges with real-world value, such as *OpenVaccine: COVID-19 mRNA Vaccine Degradation Prediction* (Das et al., 2020) and the *Vesuvius Challenge* for deciphering ancient scrolls (Lourenco et al., 2023). The total value of prizes awarded across the 75 competitions is $1,948,016 ($25,974 per competition on average).

AI agents that autonomously solve the types of challenges in our benchmark could unlock a great acceleration in scientific progress, a prospect that is exciting but also warrants careful understanding of model progress in order to deploy advancements in a safe and controlled manner. For example, MLE-bench can be used as a measure for model autonomy in OpenAI's Preparedness Framework

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